

Project Executive Proposal

SWFLN Centralized Data Reporting & Dashboard System

Prepared for: Southwest Florida Library Network (SWFLN)

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Executive Summary

Southwest Florida Library Network (SWFLN) supports member libraries through professional development, and shared resources such as events, training platforms, and equipment lending systems. Data related to these activities is currently collected across multiple third-party platforms, including calendar systems, inventory tools, training platforms, and internally maintained spreadsheets. While these systems provide export functionality, the resulting data is scattered across files and formats, making long-term reporting and analysis time-consuming and inconsistent.

This project proposes the design and development of a **lightweight, internal data aggregation and reporting system** that centralizes exported CSV and Excel data from SWFLN's existing tools into a single database. The system will provide a clear, easy-to-use dashboard that allows staff to filter data by date ranges and other criteria, and to generate **monthly reports in CSV or Excel format**.

The focus of the system is not to replace existing platforms, but to **automate everything after data export**, ensuring accuracy, consistency, and repeatability in reporting. By consolidating data into a single reporting source, SWFLN will be able to reduce manual effort, improve confidence in reported metrics, and establish a reliable foundation for long-term ROI and impact analysis.

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Project Objectives

1. **Centralize reporting data**
Store data from multiple external systems in a single, structured database to support consistent reporting over time.
2. **Improve reporting efficiency and accuracy**
Reduce manual compilation of spreadsheets by standardizing data ingestion, validation, and aggregation.
3. **Support recurring monthly reporting**
Enable staff to generate repeatable monthly reports in CSV or Excel format using predefined filters and queries.
4. **Provide a clear and intuitive dashboard**
Present data in a simple interface that supports filtering, review, and export without requiring technical expertise.
5. **Maintain a low-cost, maintainable solution**
Use open-source, lightweight technologies appropriate for an internal nonprofit reporting system.

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Data Sources & Scope

The system will ingest data exported manually by staff from the following tools:

- **Calendar / Events Platform (LibCal)**
Event and program data, including dates, attendance, and categories.
- **Inventory & Lending Platform (myTurn)**
Equipment and kit usage data, including checkouts, returns, and usage duration.
- **Career Events Data (Internal Files)**
Excel spreadsheets and invoices containing attendance, costs, dates, and vendor information.
- **Training Platform (Niche Academy)**
Training participation and completion metrics related to professional development.

All data sources currently provide exports in **CSV or Excel format**, which will be uploaded into the system.

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Implementation Plan

Phase 1: Requirements Gathering & Workflow Review

This phase focuses on understanding SWFLN's current reporting workflows and exported data formats. Meetings will be used to review example CSV/Excel files from each source, confirm required data fields, define monthly reporting requirements, and clarify how staff currently validate and use exported data. This ensures the system reflects real reporting needs and avoids unnecessary complexity.

Phase 2: System Design & Data Modeling

In this phase, the overall system architecture will be designed, including the database schema, import validation rules, and report definitions. Data from different sources will be normalized into a consistent structure to support cross-source reporting. Dashboard layouts will be planned with a focus on clarity, usability, and accuracy rather than visual branding.

Phase 3: Development, Testing, and Refinement

The final phase involves implementing the backend API, database, file ingestion pipeline, and dashboard functionality. Core features will include file uploads, validation, data storage, filtering, and CSV/Excel report generation. Testing will ensure that imported data matches expected totals and that reports are consistent month to month. Feedback from SWFLN staff will be incorporated prior to final delivery.

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Expected Benefits

1. **Single source of truth for reporting**
All reporting data is stored in one centralized system, reducing duplication and confusion.
2. **Reduced manual effort**
Staff no longer need to manually merge spreadsheets or reformat data each month.
3. **Improved reporting confidence**
Validation, import tracking, and consistent schemas ensure that reports are accurate and auditable.
4. **Scalable foundation**
The system can be expanded in the future to support additional data sources, automated ingestion, or enhanced reporting as needs evolve.

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Budget Estimate

“Hosting options and costs will be evaluated and finalized in coordination with SWFLN during the implementation phase.”

Cost Item	Description	Estimated Monthly Cost
Application Server	Hosts the web application, backend API, and MySQL database	\$0 – \$10
MySQL Database	Open-source database software (installed on server)	\$0
Backups & Storage	Disk space and basic backups	\$0 – \$5
Domain & SSL	Secure access to the system (SSL typically free)	\$0 – \$2
Estimated Total		\$0 – \$15 per month

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The system will be deployed using a standard web application architecture with a dedicated server, secure HTTPS access, persistent storage, and routine backups. This approach is appropriate for the expected scale and ensures data security and reliability.

Conclusion

This proposal outlines the development of a **centralized internal reporting system** designed to support SWFLN's monthly data reporting needs. By consolidating exported data from existing platforms into a single database and providing a clear dashboard for filtering and report generation, the system will significantly improve efficiency, accuracy, and long-term reporting consistency.

This document serves as an initial proposal to guide discussion and confirm scope. The next step is an in-person workflow review to examine real data exports, validate reporting requirements, and finalize system details prior to development.